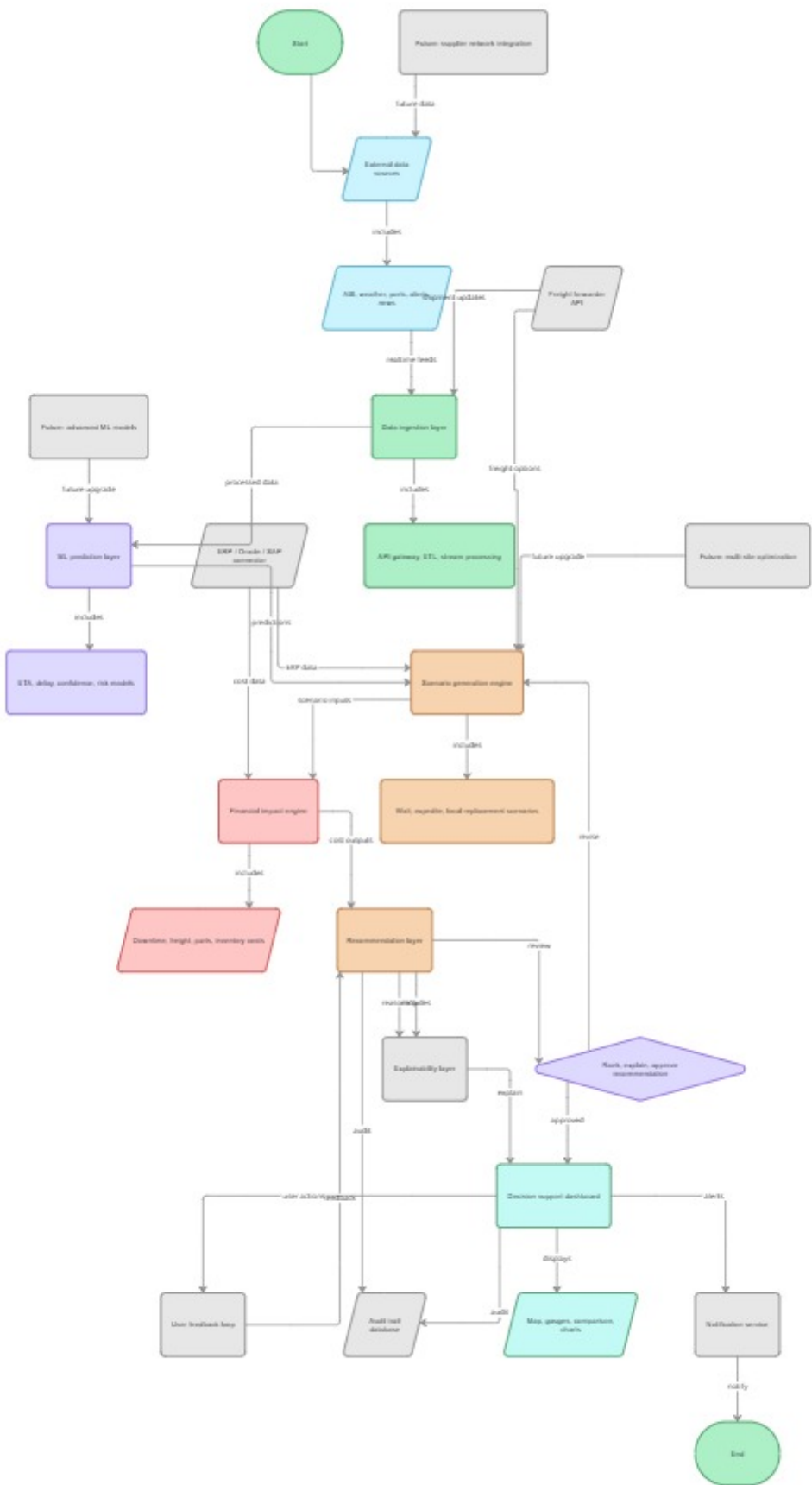








DenkKern Lena 2.0 — Living Product System Map

Comprehensive strategy
board for MVP validation
and execution

SECTION 1: Vision & Problem

From delay prediction to operational
decision support

- Not building a shipment tracking dashboard
- Help users answer: "What should we do now?"
- Convert shipment uncertainty into financial decision support

RISK: If we only show ETA and alerts, we become a fancy alerting tool

SECTION 2: Lena 2.0 Use Case

User journey:

Alert → Dashboard → Compare Scenarios → Review Recommendation → Decide → Approve

Context:

- Lena: Production manager, Hamburg shipbuilding
- Marine bolts arriving late
- Cost per delayed day: €150,000
- Choices: wait, expedite, order replacement parts

INSIGHT: Decision is about minimizing business loss, not controlling the ship

SECTION 3: Product System Flow

1. **External Signals:** AIS data, Weather, Port congestion, Maritime incidents, Strike risk, News
2. **Prediction Layer:** ETA, Delay probability, Confidence interval, Risk score
3. **Scenario Engine:** Wait | Expedite | Order replacement parts
4. **Financial Impact:** Downtime cost, Action cost, Expected loss, Total cost
5. **Decision Support:** Rank options, Explain trade-offs, Show reasoning
6. **Dashboard:** Scenario cards, Risk indicators, Recommendations, Timeline
7. **Human Decision:** Review, Select, Record

KEY: System ranks and explains. Lena decides.

Visualize a Technical Architecture — MVP

MVP Components (Build First):

- Data processing engine
- Prediction scenarios
- Manual data input (initial) for "denkkernel.ai"
- External input scenarios
- Decision support tool

Future components (Post MVP):

- Real-time integration
- User logins/API
- Dynamic scenario generation
- Automated workflows
- Knowledge layer
- Recommendation personalization

WARNING: Do not confuse the integrations with future, not MVP!

SECTION 5: Open Questions

- Q0-001: Can customers quantify cost-of-delay defensibly?
Q0-002: Who owns the cost-of-delay number?
Q0-003: Can we calibrate cost model in 1-2 weeks?
Q0-004: Is port congestion alone painful enough?
Q0-005: Can customers accept manual data intake?
Q0-006: What actions do managers actually value?
Q0-007: What explainability level is required?
Q0-008: Who is the economic buyer?
Q0-009: What data are customers unwilling to share?
Q0-010: What financial outcome justifies payment?

✓ **ACTION:** Nick validates through customer interviews

SECTION 6: Risks

- R-001: Cost-of-delay numbers may be subjective
R-002: Product becomes prediction dashboard, not decision support
R-003: Customers reject manual data intake
R-004: ERP integration too heavy for MVP
R-005: Prediction uncertainty reduces trust
R-006: Security concerns block pilots
R-007: Team overbuilds before validation

CRITICAL RISK:

Overbuilding before validation

Mitigation: Validate with customers first, build second

SECTION 7: Customer Insights

Interview Synthesis Template:

- Interview ID
- Customer role
- Pain observed
- Current workflow
- Existing tools
- Cost-of-delay estimate
- Decision options considered
- Willingness to pay
- Follow-up action

PROCESS: Every customer interview feeds this section. Build evidence incrementally.

Visualize a Technical Architecture — Future

MVP Service — Build Next:

- Data integration engine
- Data processing engine
- Prediction scenarios
- Manual data input
- "denkkernel.ai"
- External input scenarios
- Recommendation tool

Future Service — After Validation:

- Real-time integration
- User logins/API
- Dynamic scenarios
- Automated workflows
- Recommendation tool
- Knowledge layer
- Personalized recommendations

WARNING: Do not confuse the integrations with future, not MVP!







Validate before scaling.
Decision support before automation.

